INCH-POUND
MS27111B
22 February 2022
SUPERSEDING
MS27111A
1 August 1968

# DETAIL SPECIFICATION SHEET

# WASHER, KEYED, FLAT, GENERAL PURPOSE

Reactivated after 22 February 2022 and may be used for new and existing designs and acquisitions.

The requirements for acquiring the described herein shall consist of this specification sheet and specification ASME B18.21.1.

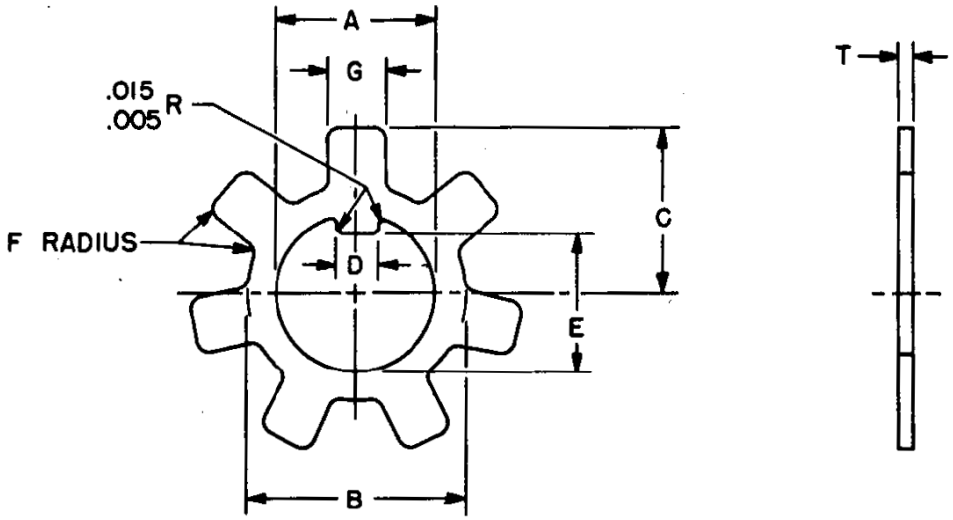



FIGURE 1. Flat washer with key.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or send email to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5310

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TABLE I. Dash numbers and dimensions.

| DASH NO. | BOLT SIZE NOM. | A ±0.005 | B ±0.010 | C ±0.010 | D ±0.005 | E ±0.005 | F ±0.005 | G ±0.010 | T | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | MAX | MIN |
| -1 | 1/4 | 0.256 | 0.374 | 0.317 | 0.055 | 0.223 | 0.032 | 0.094 | 0.033 | 0.027 |
| -2 | 5/16 | 0.319 | 0.436 | 0.364 | 0.055 | 0.286 | 0.032 | 0.116 | 0.033 | 0.027 |
| -3 | 3/8 | 0.381 | 0.499 | 0.411 | 0.086 | 0.333 | 0.032 | 0.140 | 0.033 | 0.027 |
| -4 | 7/16 | 0.444 | 0.560 | 0.458 | 0.086 | 0.396 | 0.032 | 0.164 | 0.033 | 0.027 |
| -5 | 1/2 | 0.506 | 0.684 | 0.546 | 0.117 | 0.442 | 0.032 | 0.188 | 0.033 | 0.027 |
| -6 | 9/16 | 0.569 | 0.773 | 0.626 | 0.117 | 0.505 | 0.050 | 0.210 | 0.052 | 0.044 |
| -7 | 5/8 | 0.631 | 0.896 | 0.705 | 1.148 | 0.551 | 0.050 | 0.234 | 0.052 | 0.044 |
| -8 | 3/4 | 0.756 | 1.019 | 0.798 | 0.180 | 0.661 | 0.050 | 0.280 | 0.052 | 0.044 |
| -9 | 7/8 | 0.881 | 1.206 | 0.934 | 0.180 | 0.786 | 0.050 | 0.328 | 0.052 | 0.044 |
| -10 | 1 | 1.006 | 1.367 | 1.057 | 0.242 | 0.880 | 0.062 | 0.374 | 0.065 | 0.055 |
| -11 | 1-1/8 | 1.131 | 1.517 | 1.211 | 0.242 | 1.005 | 0.062 | 0.442 | 0.065 | 0.055 |
| -12 | 1-1/4 | 1.256 | 1.689 | 1.361 | 0.305 | 1.097 | 0.062 | 0.368 | 0.065 | 0.055 |
| -13 | 1-3/8 | 1.381 | 1.880 | 1.487 | 0.305 | 1.222 | 0.062 | 0.516 | 0.065 | 0.055 |
| -14 | 1-1/2 | 1.506 | 2.029 | 1.623 | 0.367 | 1.316 | 0.078 | 0.562 | 0.081 | 0.069 |
| -15 | 1-3/4 | 1.756 | 2.392 | 1.875 | 0.430 | 1.534 | 0.078 | 0.656 | 0.081 | 0.069 |
| -16 | 2 | 2.006 | 2.754 | 2.127 | 0.492 | 1.753 | 0.078 | 0.750 | 0.081 | 0.069 |

REQUIREMENTS:

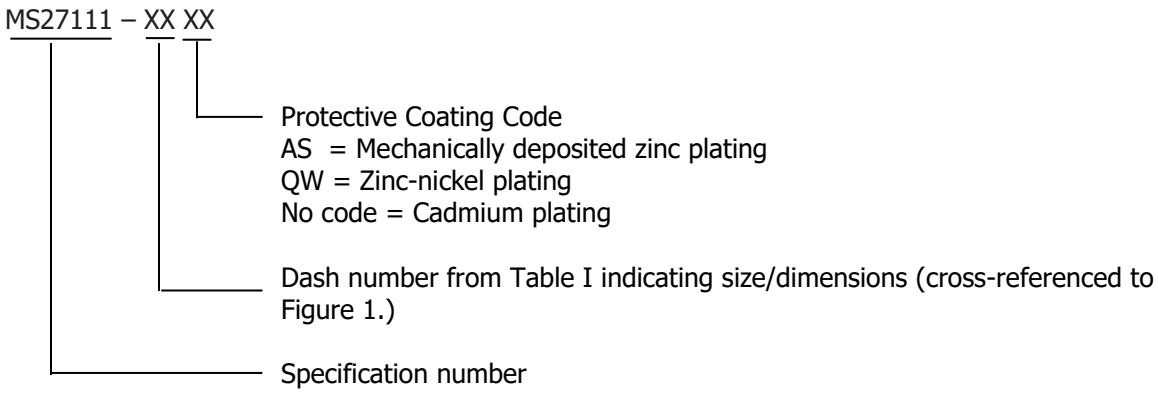
1 Steel composition. Unless otherwise specified, washers shall be made from carbon steel, SAE 1010–1035 (UNS G10100–G10350) in accordance with SAE J403.

2 Protective finish. Add one of the following codes as specified in the contract, after the dash number to designate protective finish:

AS - Zinc, Mechanically Deposited per ASTM B695, Type II, Class 5 with conversion treatment free of hexavalent chromium and meets the Type II corrosion resistance.
QW - Zinc-Nickel plate, Type K20 per MIL-PRF-32647.
No code - Cadmium plate 1/, per SAE AMS-QQ-P-416 Type II, Class 2.

1/ Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. For ground vehicle applications, both designator QW and AS are suitable alternative finishes.

3 Part or Identifying Number (PIN). The PIN shall consists of the MS number, the dash number, plus the finish code.




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PIN examples: MS27111-1AS is a mechanically deposited zinc finished, carbon steel, flat washer with key, and having a nominal 1/4 inch bolt-hole diameter.

MS27111-5QW is a zinc-nickel plated, carbon steel, flat washer with key, and having a nominal 1/2 inch bolt-hole diameter.

MS27111-9 is a cadmium plated, carbon steel, flat washer with key, and having a nominal 7/8 inch bolt-hole diameter.

4 Dimensions. Unless otherwise specified, all dimensions are in inches.

5 Edges and burrs. Break sharp edges and remove all burrs.

6 Flatness. Washers shall be flat to within 0.010 inch.

7 Design. For design feature purposes, this document takes precedence over all other documents.

8 Lugs. Lugs shall be equally spaced. One lug shall be located on axis of key.

9 Documents. Referenced documents shall be of the issue in effect on the date of the invitation for bids unless otherwise specified in the contract. Copies of military specifications are available online at https://quicksearch.dla.mil. Copies of ASME documents are available online at https://www.asme.org. Copies of SAE documents are available online at https://www.sae.org/standards.

10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – AS
Air Force – 71
DLA – GS

Review activities:
Army – AR, CR4, MI
Navy – MC, OS

Preparing Activity:
Army – AT

(Project 5310-2022-007)

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